

Clustering of multivariate tail dependence using conditional methods

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UNIVERSITY OF
BATH



**UK Research
and Innovation**

Introduction



Storm Darragh causes blackouts and travel disruption



Problem

- Of interest:

$$\mathbb{P}(Y_1 > y_1, Y_2 > y_2) = \mathbb{P}(Y_1 > y_1) \mathbb{P}(Y_2 > y_2 \mid Y_1 > y_1)$$

for large y_1, y_2

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- “concomitant” extremes particularly devastating
- Goal: identify trends by clustering sites with similar **tail dependence**

Setting

- Observe d variables at sites $s = 1, \dots, D$
- Estimate **tail dependence** at each site
- Cluster sites with similar tail dependence

Dependence

Asymptotic dependence

- Coefficient of extremal dependence $\chi \in [0, 1]$,

$$\chi = \lim_{u \rightarrow 1} \mathbb{P}[F_1(Y_1) > u \mid F_2(Y_2) > u]$$

- (Increasingly strong) asymptotic dependence for $\chi > 0$

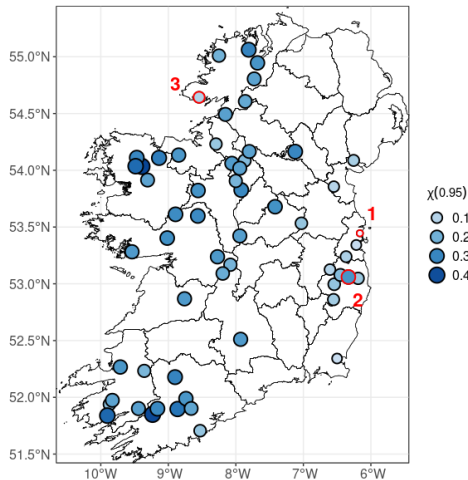
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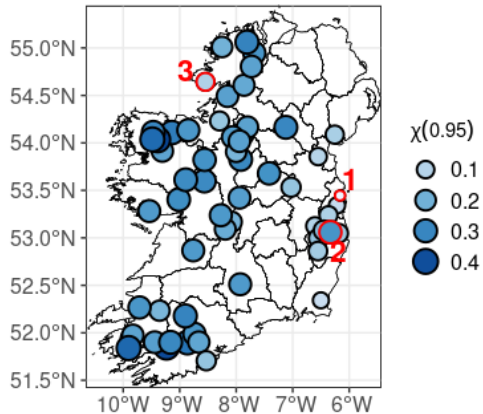
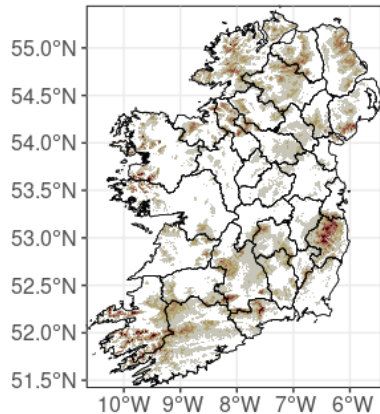
- (Increasingly strong) asymptotic dependence for $\chi > 0$
- However, χ only gives summary; inference requires **dependence model**

Ireland



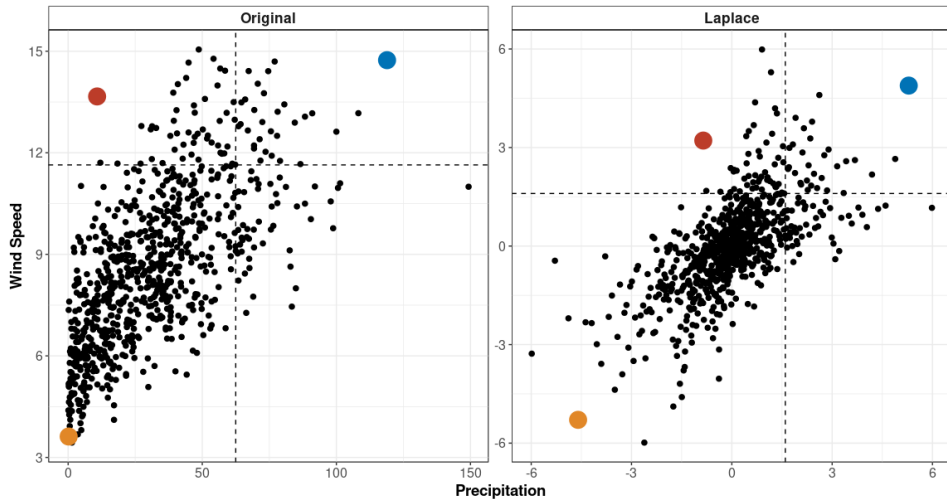
Data from Met Éireann, ERA5 reanalysis

Ireland



Conditional extremes

Marginal transformation



Conditional extremes

- Heteroscedastic regression dependence model:

$$(Y_2 \mid Y_1 = y) = \alpha y + y^\beta Z, \text{ for } y > u$$

- slope parameter α for Y_2 given large Y_1 ,
- $\beta \in (-\infty, 1]$ controls “spread” induced by large Y_1

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- **Key assumption:** Residuals $Z \sim N(\mu, \sigma)$

Conditional extremes

- Heteroscedastic regression dependence model:

$$(Y_{-i} \mid Y_i = y) = \alpha_{|i}y + y^{\beta_{|i}}Z_{|i}, \text{ for } y > u_i$$

- $Y_{-i} := Y \setminus Y_i \in \mathbb{R}^{d-1}$
 - slope parameter $\alpha_{j|i} \in [-1, 1]$ for Y_j given large Y_i
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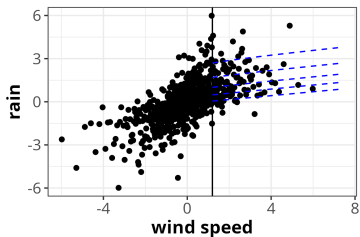
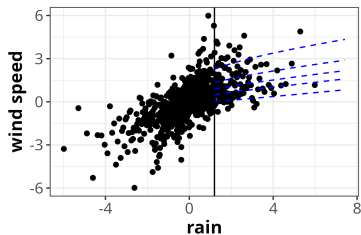
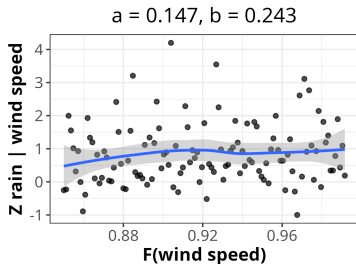
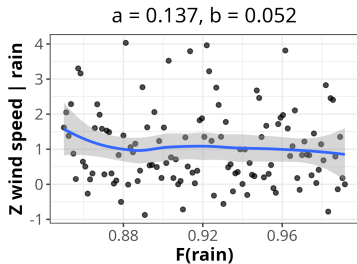
- Heteroscedastic regression dependence model:

$$(Y_{-i,s} \mid Y_{i,s} = y) = \alpha_{|i,s} y + y^{\beta_{|i,s}} Z_{|i,s}, \text{ for } y > u_{i,s}$$

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Conditional extremes

Costelloe Fishery



Inference

Inference assumes conditional distribution follows (MVN) distribution:

$$(\mathbf{Y}_{-i,s} \mid Y_{i,s} = y) \sim \text{MVN}_{d-1} \left(\boldsymbol{\alpha}_{|i,s} y + y^{\boldsymbol{\beta}_{|i,s}} \boldsymbol{\mu}_{|i,s}, (y^{\boldsymbol{\beta}_{|i,s}})^{\text{T}} \boldsymbol{\Sigma}_{|i,s} y^{\boldsymbol{\beta}_{|i,s}} \right)$$

where $y > u_{i,s}$

$\implies$ comparing dependence structures at different sites
equivalent to comparing MVN distributions

Clustering

Kullback-Leibler divergence

- KL between two continuous $(d - 1)$ -variate distribution functions H and H^* is

$$\text{KL} (H \| H^*) = \int_{\mathbb{R}^{d-1}} h(\mathbf{y}) \log \left(\frac{h(\mathbf{y})}{h^*(\mathbf{y})} \right) d\mathbf{y},$$

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- $\text{KL}(H \| H^*)$ has closed form when H, H^* are MVN
- $\text{KL} (H \| H^*) + \text{KL} (H^* \| H)$ symmetric but can be unstable

Skew-geometric Jensen-Shannon divergence

- JS^{G_λ} between H, H^* is

$$\begin{aligned}\text{JS}^{G_\lambda}(H \| H^*) &= (1 - \lambda) \text{KL} \{G_\lambda(H, H^*) \| H\} \\ &\quad + \lambda \text{KL} \{G_\lambda(H, H^*) \| H^*\}\end{aligned}$$

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- $G_\lambda(H, H^*) = H^{1-\lambda}(H^*)^\lambda$ is geometric mean with $\lambda \in [0, 1]$
- Symmetric, closed form, more stable than KL

Conditional $\text{JS}^{\text{G}_\lambda}$

- $\text{JS}^{\text{G}_\lambda}$ for sites s, s^* *conditional* on variable i :

$$\text{cJS}_{i,s,s^*}^{\text{G}_\lambda}(y) = \text{JS}^{\text{G}_\lambda} (H_{i,s}(y) \| H_{i,s^*}(y)) , \quad y > u_i,$$

where $H_{i,s}(y), H_{i,s^*}(y)$ are conditional MVNs at s, s^*

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- $\text{cJS}_{i,s,s^*}^{\text{G}_\lambda}$ depends on choice of $y > u_i \dots$

Expected $cJS_{i,s,s^*}^{G_\lambda}$

$$\begin{aligned} eJS_{i,s,s^*}^{G_\lambda} &= \mathbb{E} \left(cJS_{i,s,s^*}^{G_\lambda}(Y_{i,s}) \mid Y_{i,s} > u_i \right) \\ &= \int_{u_i}^{\infty} cJS_{i,s,s^*}^{G_\lambda}(y) h(y \mid Y_{i,s} > u_i) dy \end{aligned}$$

where $h(y \mid Y_{i,s} > u_i)$ is standard Laplace

Dissimilarity matrix

- For each conditioning variable $i \in \{1, \dots, d\}$, compute pairwise $\text{eJS}^{\text{G}_\lambda}$ dissimilarities between all D sites:

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- Can aggregate over all d conditioning variables to get overall dissimilarity matrix:

$$M = \frac{1}{d} \sum_{i=1}^d M^{(i)}$$

Clustering

- Use **k-medoids** to cluster over M (or $M^{(i)}$)
- Use standard methods to choose k

Simulations

Gaussian copula

- 10,000 bivariate observations at $D = 12$ “sites”
- Bivariate Gaussian copula with correlation $\rho_{\text{Gauss}}^s \in [0, 1]$ at site $s = 1, \dots, D$

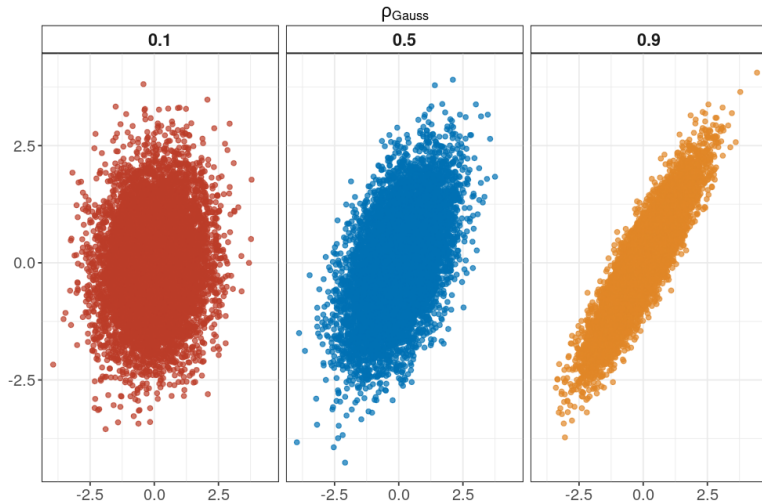
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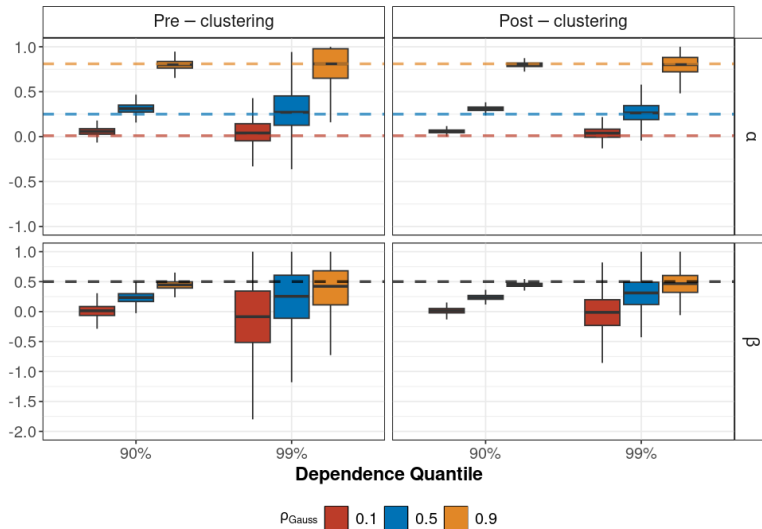
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- Example: 3 clusters of 4 locations defined by respective ρ_{Gauss}^s values of 0.1, 0.5, 0.9

Gaussian copula



Gaussian copula - results



Mixture simulation

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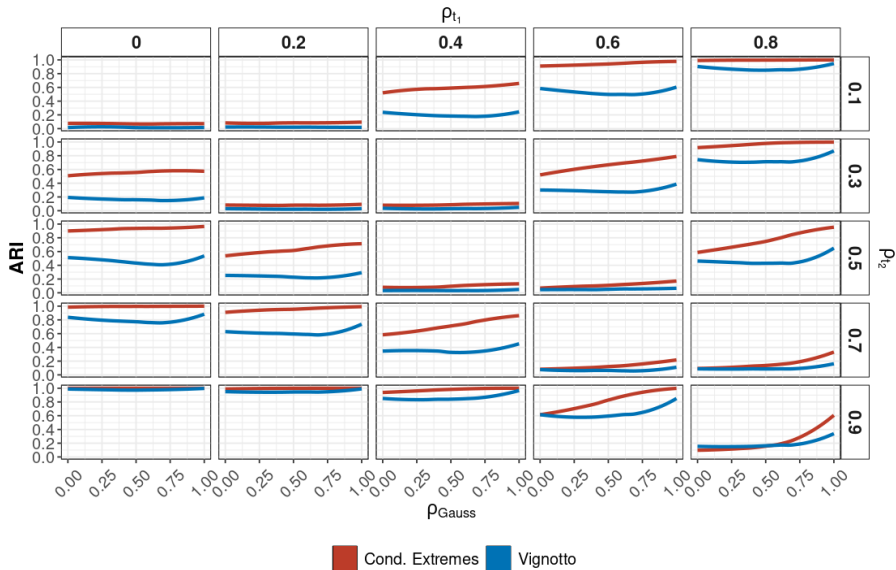
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- Compared to competing Vignotto algorithm

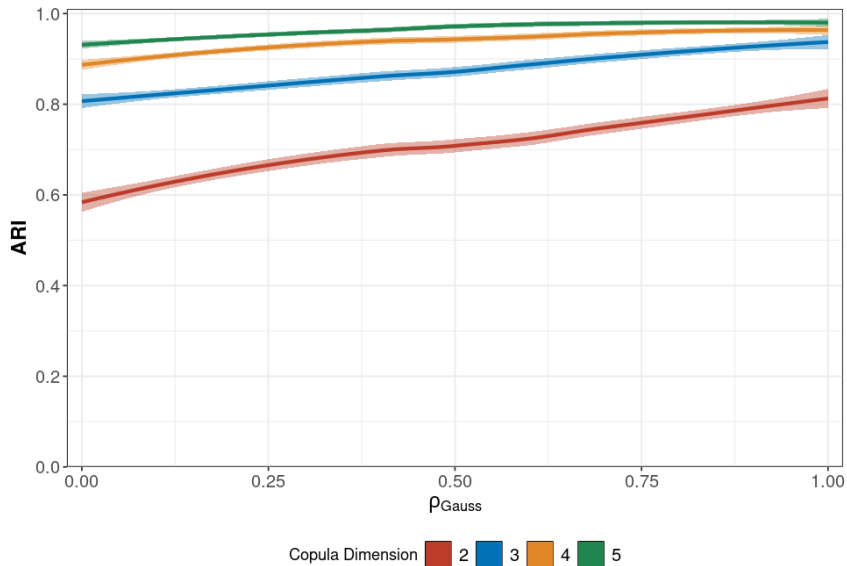
Vignotto, Engelke, and Zscheischler,

“Clustering bivariate dependencies of compound precipitation and wind extremes over Great Britain and Ireland” (2021)

Mixture simulation - results

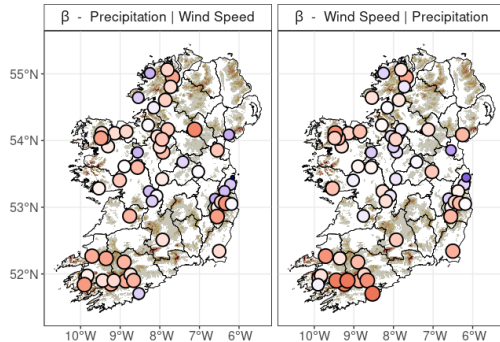
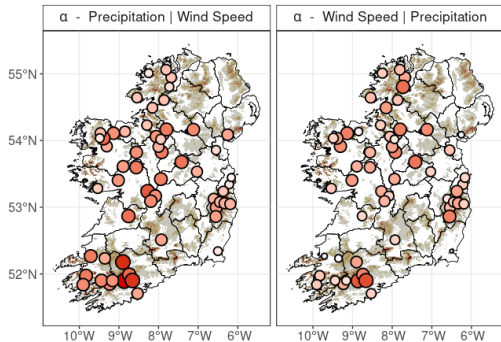


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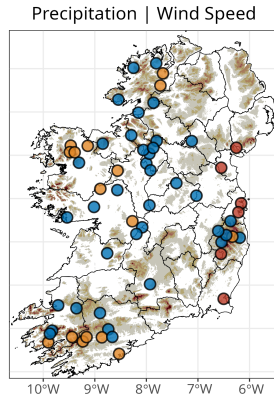
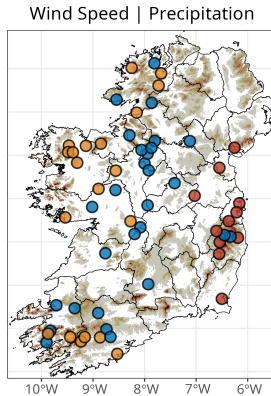
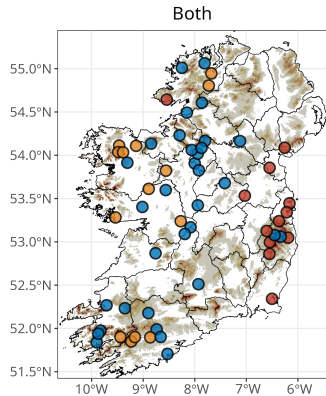


Ireland meteorological data

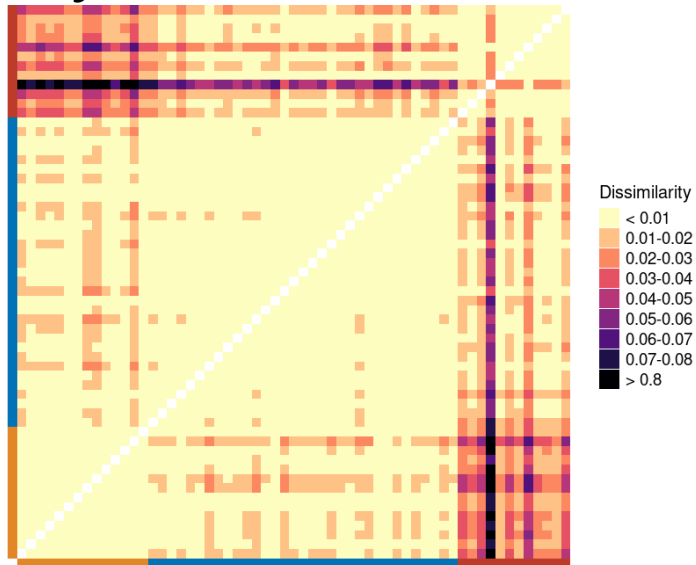
CE estimates



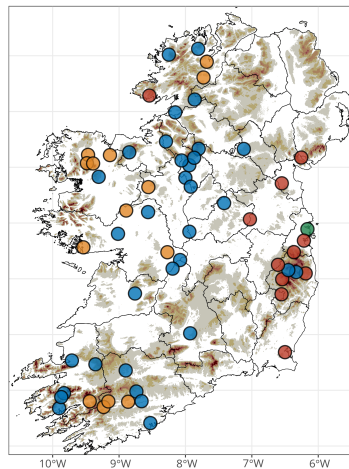
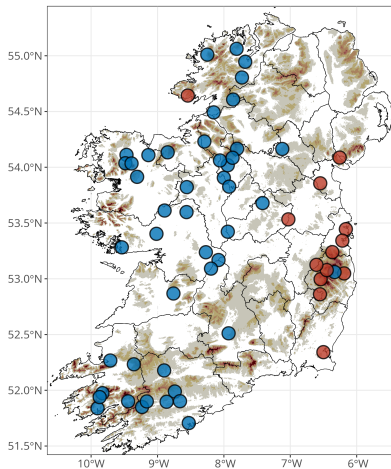
Clustering



Dissimilarity matrix



$k = 2$ & $k = 4$



Discussion

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- **Conclusion:**
 - Principled & effective clustering framework for CE models
 - Simulations & application show clustering can group sites with similar tail dependence, which may aid interpretation

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- **Conclusion:**
 - Principled & effective clustering framework for CE models
 - Simulations & application show clustering can group sites with similar tail dependence, which may aid interpretation
- **Limitations:**
 - **CE:** Gaussian assumption for $Z_{|i,s}$
 - **Clustering:** Uncertainty in CE fits is ignored in clustering

Thank You!



<https://arxiv.org/abs/2510.20424>



<https://github.com/potoole7/CeCl>

References I

-  Coles, Stuart, Janet Heffernan, and Jonathan Tawn (1999). “Dependence measures for extreme value analyses”. In: *Extremes* 2.4, pp. 339–365.
-  Heffernan, Janet E and Jonathan A Tawn (2004). “A conditional approach for multivariate extreme values (with discussion)”. In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 66.3, pp. 497–546.

References II

-  Nielsen, Frank (May 2019). “On the Jensen-Shannon Symmetrization of Distances Relying on Abstract Means”. In: *Entropy* 21.5, p. 485. ISSN: 1099-4300. DOI: 10.3390/e21050485. URL: <http://dx.doi.org/10.3390/e21050485>.
-  Vignotto, Edoardo, Sebastian Engelke, and Jakob Zscheischler (2021). “Clustering bivariate dependencies of compound precipitation and wind extremes over Great Britain and Ireland”. In: *Weather and Climate Extremes* 32, p. 100318.

Appendix

$\text{JS}^{\text{G}_\lambda}$ **closed form for** MVNs

- For $H \sim \text{MVN}(\mathbf{m}, \Omega)$, $H^* \sim \text{MVN}(\mathbf{m}^*, \Omega^*)$,

$$\text{JS}^{\text{G}_\lambda}(H \| H^*) = \frac{1}{2} \left[(1 - \lambda) \mathbf{m}^T \Omega^{-1} \mathbf{m} + \lambda (\mathbf{m}^*)^T (\Omega^*)^{-1} \mathbf{m}^* \right. \\ \left. - \mathbf{m}_\lambda^T \Omega_\lambda^{-1} \mathbf{m}_\lambda + \log \left(\frac{|\Omega|^{1-\lambda} |\Omega^*|^\lambda}{|\Omega_\lambda|} \right) \right],$$

$$\mathbf{m}_\lambda = \Omega_\lambda \left[(1 - \lambda) \Omega^{-1} \mathbf{m} + \lambda (\Omega^*)^{-1} \mathbf{m}^* \right],$$

$$\Omega_\lambda = \left[(1 - \lambda) \Omega^{-1} + \lambda (\Omega^*)^{-1} \right]^{-1}$$

- Ω_λ is **harmonic matrix barycentre** of Ω and Ω^*
- $\text{JS}^{\text{G}_\lambda}(H \| H^*) \leq \frac{1}{2} \max\{\lambda, 1 - \lambda\} [\text{KL}(H \| H^*) + \text{KL}(H^* \| H)]$

Clustering

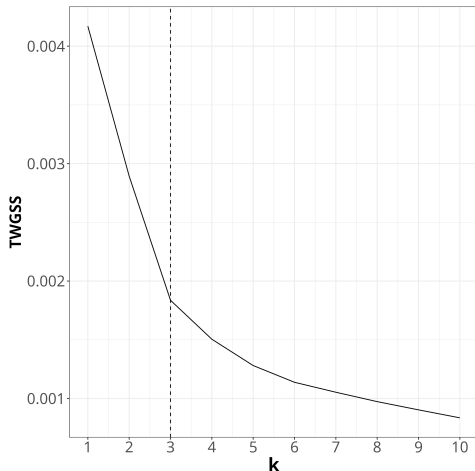
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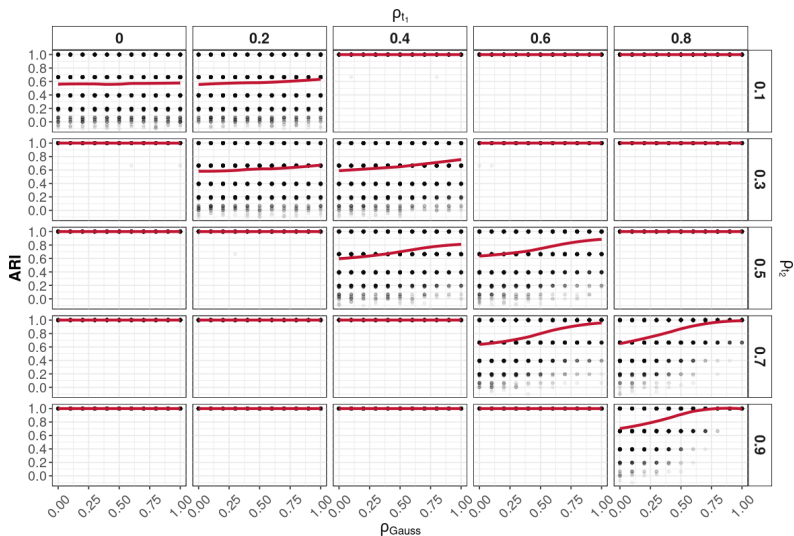
- Use **k-medoids** to cluster over M (or $M^{(i)}$)
- **Elbow plot:** k chosen using **Total Within Group Sum of Squares**:

$$\text{TWGSS}(k) = \sum_{c=1}^k \sum_{s \in \mathcal{C}_c} \text{eJS}_{i,s,\kappa_c}^{G_\lambda},$$

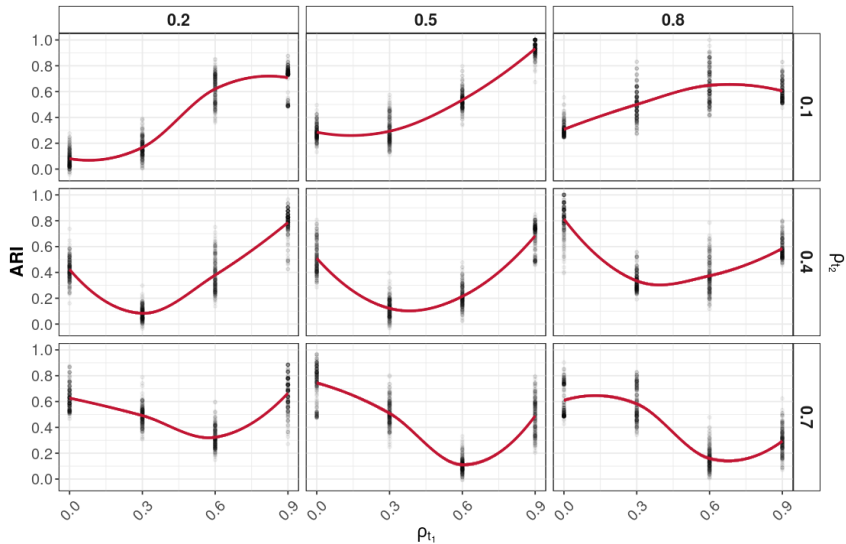
where $\mathcal{C}_c$, κ_c are cluster c and its medoid



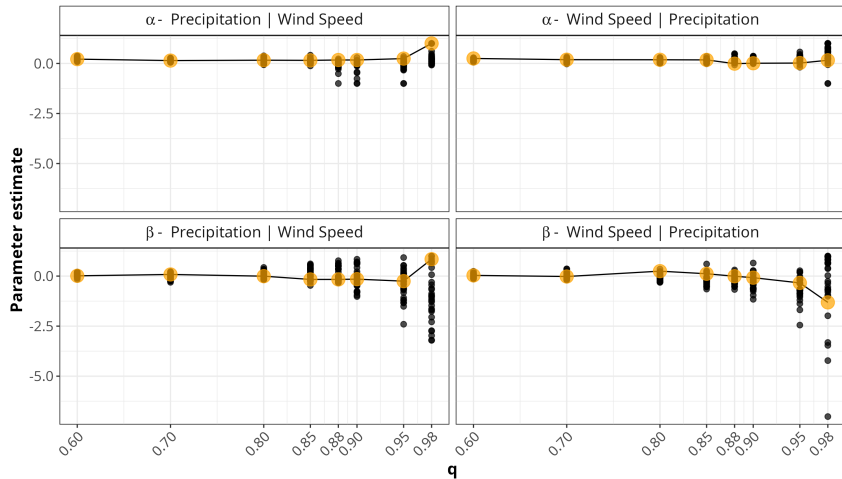
Mixture simulation (three variables)



More realistic example



Threshold stability



Sensitivity analysis

